

1. Executive Summary

		Traffic Summary		Count	
		Total Flows Observed		15580	
		Total Packets Analyzed		4505359661	
		Malicious Packets		4495136719	
		True Negatives Passed		10922	
		Attack Vector		Count	
		ICMP Flood		0	
		SYN Flood		0	
		UDP Flood		9	
		Uncertain		3	
		Action		Count	
		Quarantined		3	
		Time Ban		3	
		Blackhole		0	
		Released		0	

Metric		Value	
Total Threats Mitigated		12	
High Priority		9	
Low Priority		3	
Manual Releases		0	
Manual Blocks		0	
ML Metrics		Value	
False Positives		0	
FP Rate		0.00%	
IF Precision		100.00%	
IF Recall		97.73%	
IF F1-Score		98.85%	
IF Accuracy		99.31%	

2. Performance Benchmark

2a. Isolation Forest: Anomaly Detection

Metric	Value	Description
Precision	100.00%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	97.73%	Share of actual attacks successfully flagged
F1-Score	98.85%	Balanced measure combining Precision and Recall
Accuracy	99.31%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.00%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	2.27%	Actual attacks that went undetected
True Positive Rate (TPR)	97.73%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	100.00%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 4652	FN 108
Actual: Normal	FP 0	TN 10814

2b. Random Forest: Attack Classification

Metric	Value	Description
Precision	100.00%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	100.00%	Share of actual attacks successfully flagged
F1-Score	100.00%	Balanced measure combining Precision and Recall
Accuracy	100.00%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.00%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	0.00%	Actual attacks that went undetected
True Positive Rate (TPR)	100.00%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	100.00%	Normal traffic correctly identified as safe

	Predicted: SYN	Predicted: ICMP	Predicted: UDP
Act: SYN	0	0	0
Act: ICMP	0	0	0
Act: UDP	0	0	4652

2c. Hybrid Cascaded

Metric	Value	Description
Precision	100.00%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	97.73%	Share of actual attacks successfully flagged
F1-Score	98.85%	Balanced measure combining Precision and Recall
Accuracy	99.31%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.00%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	2.27%	Actual attacks that went undetected
True Positive Rate (TPR)	97.73%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	100.00%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 4652	FN 108
Actual: Normal	FP 0	TN 10814

2d. Response Latency

Metric	Value	Description
Detection Time	744.2 ms	Average time taken to detect an attack
Mitigation Response Time	0.5 ms	Interval between the anomaly flag and the FlowMod blocking rule install

2e. Controller Resource Overhead

Metric	Value	Description
Controller CPU (Baseline)	1.20%	Ryu controller CPU usage under normal traffic
Controller CPU (Active Attack)	6.52%	Ryu controller average CPU usage during a DDoS simulation
Controller CPU (Mitigation)	6.52%	Ryu controller CPU usage during simultaneous detection and mitigation

3. Offences Summary

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.20	4	3	2026-09-26 18:39:49	2026-09-26 18:42:30
10.0.0.21	4	3	2026-09-26 18:39:55	2026-09-26 18:42:34
10.0.0.22	4	3	2026-09-26 18:39:48	2026-09-26 18:42:29

4. Chronological Mitigation Log

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-26 18:39:48	10.0.0.22	DDoS	Uncertain	-	Low	Quarantined
2026-09-26 18:39:49	10.0.0.20	DDoS	Uncertain	-	Low	Quarantined
2026-09-26 18:39:55	10.0.0.21	DDoS	Uncertain	-	Low	Quarantined
2026-09-26 18:39:59	10.0.0.22	DDoS	UDP Flood	76.5%	High	Time Ban (30s)
2026-09-26 18:40:00	10.0.0.20	DDoS	UDP Flood	77.7%	High	Time Ban (30s)
2026-09-26 18:40:01	10.0.0.20	DDoS	UDP Flood	77.7%	High	Time Ban
2026-09-26 18:40:03	10.0.0.22	DDoS	UDP Flood	76.5%	High	Time Ban
2026-09-26 18:40:05	10.0.0.21	DDoS	UDP Flood	78.1%	High	Time Ban (30s)
2026-09-26 18:40:06	10.0.0.21	DDoS	UDP Flood	78.1%	High	Time Ban
2026-09-26 18:41:00	10.0.0.22	DDoS	UDP Flood	79.1%	High	Time Ban (1m)
2026-09-26 18:41:01	10.0.0.20	DDoS	UDP Flood	79.1%	High	Time Ban (1m)
2026-09-26 18:41:06	10.0.0.21	DDoS	UDP Flood	79.5%	High	Time Ban (1m)

5. IP Attack History (Completed Sessions)

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.21	UDP Flood	0.6305	77.7%	Critical	Phase 2	1	11s	2026-09-26 18:42:34	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6329	77.9%	Critical	Phase 2	1	11s	2026-09-26 18:42:30	Attack persisted - escalating to time ban
10.0.0.22	UDP Flood	0.6398	75.0%	Critical	Phase 2	1	11s	2026-09-26 18:42:29	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6147	79.5%	High	Phase 2	3	1m 11s	2026-09-26 18:41:06	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6284	79.1%	High	Phase 2	3	1m 12s	2026-09-26 18:41:01	Attack persisted - escalating to time ban
10.0.0.22	UDP Flood	0.6231	79.1%	High	Phase 2	3	1m 12s	2026-09-26 18:41:00	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6079	79.3%	High	Phase 2	2	41s	2026-09-26 18:40:36	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6309	77.7%	High	Phase 2	2	42s	2026-09-26 18:40:31	Attack persisted - escalating to time ban
10.0.0.22	UDP Flood	0.6402	76.5%	High	Phase 2	2	42s	2026-09-26 18:40:30	Attack persisted - escalating to time ban

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.21	UDP Flood	0.6316	78.1%	High	Phase 2	1	10s	2026-09-26 18:40:05	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6329	77.7%	High	Phase 2	1	11s	2026-09-26 18:40:00	Attack persisted - escalating to time ban
10.0.0.22	UDP Flood	0.6398	76.5%	High	Phase 2	1	11s	2026-09-26 18:39:59	Attack persisted - escalating to time ban

Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

SIGNATURE OVER PRINTED NAME
Network Administrator

DATE